

Executive Summary:

Host Vocabulary Predicts Whether New Scientific Concepts Take Root Across Disciplinary Boundaries

Internal Research Report

September 2026

Headline

When a scientific concept enters a new disciplinary subfield, the share of its initial co-occurrence partners that are native to that host subfield predicts whether the concept will subsequently be taken up by newcomer authors. This host-entry grafting effect was confirmed on the sealed held-out fold (co-primary IRR/SD 1.19 [1.06, 1.33], $p = 0.004$) and replicated on an independent MeSH biomedical population for biomedicine-to-biomedicine host entries (co-primary IRR/SD 1.23 [1.12, 1.36], Holm $p < 0.001$). The IVW pool of the main-arm screen and MeSH co-primary estimates is 1.26 [1.17, 1.36] with $I^2 = 0$. The effect is on the count of newcomer papers; binary establishment is null on held-out. The share of origin companions (co-transfer) is null everywhere.

Goal

The study investigates how emerging scientific concepts spread across disciplinary boundaries, using 426 OpenAlex concepts (366 emerging, 60 stationary reference; 462,812 works) and 191 new MeSH biomedical concepts. Two research questions guide the work:

- **RQ1 (Emergence):** Which temporal and structural patterns in an evolving knowledge graph characterise concept emergence?
- **RQ2 (Diffusion):** How do emerging concepts diffuse across disciplinary communities, and what distinguishes locally concentrated concepts from those that become broadly integrated?

The target venue is Applied Network Science. The study ran five iterations over a pre-registered, frozen concept pool.

What Was Tried

Iteration 1: Data foundation and pre-registration. Built a 184-concept pool (later expanded to 426), constructed yearly co-occurrence networks, assembled a prior-art dossier ranking five candidate mechanisms by novelty, and froze a pre-registered screen with decision rules.

Iteration 2: Citation gate, event study, MeSH replication. Attempted a citation-layer viability decomposition (Gate A). Only 17.3% of edges met the within-host tracing threshold; Gate A failed and triggered the pre-registered graft fallback. Ran the RQ1 closure event study on the screen fold (its held-out test followed in iteration 4). Built yearly co-word snapshots ($\approx 27,000$ nodes, $\approx 84,000$ edges) and ran the precursor event study.

Iteration 3: Deepening closure and testing grafting. Decomposed the closure effect into turnover, brokerage, and community components. Tested whether openness predicts breadth (not supported). Ran the host-entry grafting test: co-primary specification IRR/SD 1.30 [1.16, 1.45], $p = 1 \times 10^{-5}$, on the screen fold. Ran a descriptive diffusion typology yielding a stable $k = 2$ solution: localised ($n = 136$) vs broad-from-the-start ($n = 66$).

Iteration 4: Held-out confirmation and cross-population replication. Opened the sealed held-out folds. RQ1 is R1_DEAD under the frozen kill rule (raw closure Holm $p = 0.147$): structural precursors of sustained uptake are volume/churn correlates; persistent-neighbour closure is the one confirmed row (pooled-panel coefficient -1.07 , Holm $p = 0.015$). Grafting confirmed on held-out (IRR/SD 1.19 [1.06, 1.33]) and replicated on MeSH, biomedicine-to-biomedicine entries only (IRR/SD 1.23 [1.12, 1.36]). Four discriminating tests passed: single-paper artefact (ratio 0.80, $p = 0.004$), vocabulary decomposition (NATIVE and ADJACENT both positive), topic-score absorption (-7.3% to -10.3% , not meaningful), and closure-anchoring link (null, $\rho = 0.049$, confirming independence from closure).

Iteration 5: Post-confirmation decomposition and audit. Decomposed the grafting effect along three dimensions: extensive vs intensive margin (verdict: BOTH, extensive margin 38% of total log-IRR), host-specificity (verdict: HOST-SPECIFIC, A_{cont} retention 0.969), and field boundary (verdict: no boundary, Wald $p = 0.638$). Ran a full numerical audit (11 modules, 99.4% curated-number accuracy, all 6 reviewer critiques closed). Produced a positioning dossier (PARTLY ANTICIPATED) and publication-ready figure specifications.

Key Results

1. **Grafting effect confirmed and replicated.** A one-SD increase in host-nativeness of entry partners yields $\approx 19\text{--}30\%$ more newcomer uptake. IVW pool of screen and MeSH IRR/SD 1.26 [1.17, 1.36]. The mechanism label is “host-vocabulary.”
2. **Effect is host-specific, not generic.** Generality controls leave A_{cont} unchanged (retention 0.969). The Balassa-index A_{lft} replicates across all folds (IVW IRR/SD 1.34 [1.23, 1.46]).
3. **Both margins matter.** Host-leaning entry vocabulary raises the probability of any newcomer uptake (extensive, $+6.43$ pp/SD) and its magnitude given uptake (intensive, IRR/SD 1.183).
4. **No detectable field boundary.** Across origin-field groups (Physics/Astronomy, Computer Science, other) on the pooled screen and held-out folds, the Wald test of equality is null ($p = 0.638$); the Physics/Astronomy group alone is non-significant (IRR/SD 1.18 [0.93, 1.50]).
5. **RQ1 is R1_DEAD.** Structural precursors of sustained uptake are volume/churn correlates (primary closure Holm $p = 0.147$). Qualification: persistent-neighbour closure is confirmed on held-out (pooled-panel coefficient -1.07 , Holm $p = 0.015$), though its screen event-study interval included zero. Burt constraint is higher, so brokerage is ruled out.
6. **Diffusion typology ($k = 2$).** Localised (136 concepts, low entropy, physics-dominated) vs broad-from-the-start (66 concepts, high entropy, mixed origin).
7. **Adopter mechanism (screen only).** Adopters have higher odds of prior exposure to entry partners (OR 3.09 [2.32, 4.28]; prevalence 0.79 vs 0.61, risk ratio about 1.3). Enrichment is strongest for FOREIGN partners (OR 2.88); consistent with absorptive capacity or topical proximity.
8. **Expansion precedes diffusion, conditionally.** In 25 of 26 pooled dual-onset main-arm concepts (8.6% of 302), but diffusion onsets are rarer overall and the MeSH ordering (9 of 13) is not different from a year-shuffle null ($p = 0.43$).

What Failed

1. **Gate A (citation-layer viability).** Within-host traced share 0.204 vs 0.477 any-parent. The viability decomposition central to the original hypothesis cannot be executed.
2. **Primary closure measure (R1_DEAD).** Holm $p = 0.147$ under the frozen kill rule.
3. **Precursors add no predictive value** for WHETHER a concept emerges (E_{up} logit dAUC -0.010 $[-0.045, 0.021]$).
4. **Breadth prediction not supported.** Openness does not predict where concepts travel (all Holm $p = 0.405$).
5. **Co-transfer is null** everywhere: origin companions do not predict newcomer uptake.
6. **Community roles do not predict host entry.** BRIDGE OR 0.64 $[0.38, 1.08]$.
7. **3-channel diffusion typology** adds nothing beyond entropy.
8. **Binary establishment null on held-out.** EST_bin $p = 0.16$.
9. **Grafting primary specification inconclusive** on held-out (IRR/SD 0.98 $[0.70, 1.37]$, 30 clusters).
10. **292 MISSING_IN_REPORT rows** in the numerical audit—numbers in artifact files not transcribed.

Open Questions

1. **Causal identification.** The grafting result is associational. Concepts entering with host-native partners may differ on unobserved dimensions.
2. **Generalisation beyond physics/CS/biomedicine.** The main-arm pool is dominated by arXiv-sourced concepts; social sciences and humanities are untested.
3. **Binary establishment.** The strict threshold (≥ 3 newcomer papers and host papers in ≥ 3 of 5 years) is null on the held-out fold. The claim holds for uptake initiation ($Y \geq 1$) but not for sustained establishment.
4. **Single-paper artefact.** 87% of co-primary events are single-paper entries; the held-out interaction ($p = 0.004$) shows single-paper entries drive the effect more strongly.
5. **Intensive margin is conditional.** The margin decomposition conditions on $Y \geq 1$, introducing selection bias.
6. **Adopter-level mechanism ambiguity.** The design cannot separate absorptive capacity from topical proximity.
7. **Seeded-perturbation recall 0.60.** Estimator/fold relabelling detection at 0.10.